Solution

Let P, Q, R be the points on the line segment joining $A(-11,4)$ and $B(9,8)$ such that $AP = PQ = QR = RB$.

Here Q is the mid-point of AB, P is the mid-point of AQ and R is the mid-point of QB.

$$Q \text{ is the mid-point of } AB = \left(\frac{-11+9}{2}, \frac{4+8}{2}\right) = \left(\frac{-2}{2}, \frac{12}{2}\right) = (-1,6)$$

$$P \text{ is the mid-point of } AQ = \left(\frac{-11-1}{2}, \frac{4+6}{2}\right) = \left(\frac{-12}{2}, \frac{10}{2}\right) = (-6,5)$$

$$R \text{ is the mid-point of } QB = \left(\frac{-1+9}{2}, \frac{6+8}{2}\right) = \left(\frac{8}{2}, \frac{14}{2}\right) = (4,7)$$

Hence the points which divides AB into four equal parts are $P(-6, 5)$, $Q(-1, 6)$ and $R(4, 7)$.

Example 5.15 The mid-points of the sides of a triangle are $(5,1)$, $(3,-5)$ and $(-5,-1)$. Find the coordinates of the vertices of the triangle.

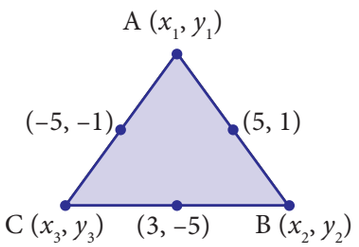


Fig. 5.27

Solution Let the vertices of the $\varnothing ABC$ be $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ and the given mid-points of the sides AB, BC and CA are $(5,1)$, $(3,-5)$ and $(-5,-1)$ respectively. Therefore,

$$\frac{x_1 + x_2}{2} = 5 \Rightarrow x_1 + x_2 = 10 \quad ...(1)$$

$$\frac{x_2 + x_3}{2} = 3 \Rightarrow x_2 + x_3 = 6 \quad ...(2)$$

$$\frac{x_3 + x_1}{2} = -5 \Rightarrow x_3 + x_1 = -10 \quad ...(3)$$

Adding (1), (2) and (3)

$$2x_1 + 2x_2 + 2x_3 = 6$$

$$x_1 + x_2 + x_3 = 3 \quad ...(4)$$

$(4) - (2) \Rightarrow x_1 = 3 - 6 = -3$

$(4) - (3) \Rightarrow x_2 = 3 + 10 = 13$

$(4) - (1) \Rightarrow x_3 = 3 - 10 = -7$

$$\frac{y_1 + y_2}{2} = 1 \Rightarrow y_1 + y_2 = 2 \quad ...(5)$$

$$\frac{y_2 + y_3}{2} = -5 \Rightarrow y_2 + y_3 = -10 \quad ...(6)$$

$$\frac{y_3 + y_1}{2} = -1 \Rightarrow y_3 + y_1 = -2 \quad ...(7)$$

Adding (5), (6) and (7),

$$2y_1 + 2y_2 + 2y_3 = -10$$

$$y_1 + y_2 + y_3 = -5 \quad ...(8)$$

$(8) - (6) \Rightarrow y_1 = -5 + 10 = 5$

$(8) - (7) \Rightarrow y_2 = -5 + 2 = -3$

$(8) - (5) \Rightarrow y_3 = -5 - 2 = -7$

Therefore the vertices of the triangles are $A(-3,5)$, $B(13,-3)$ and $C(-7,-7)$.

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